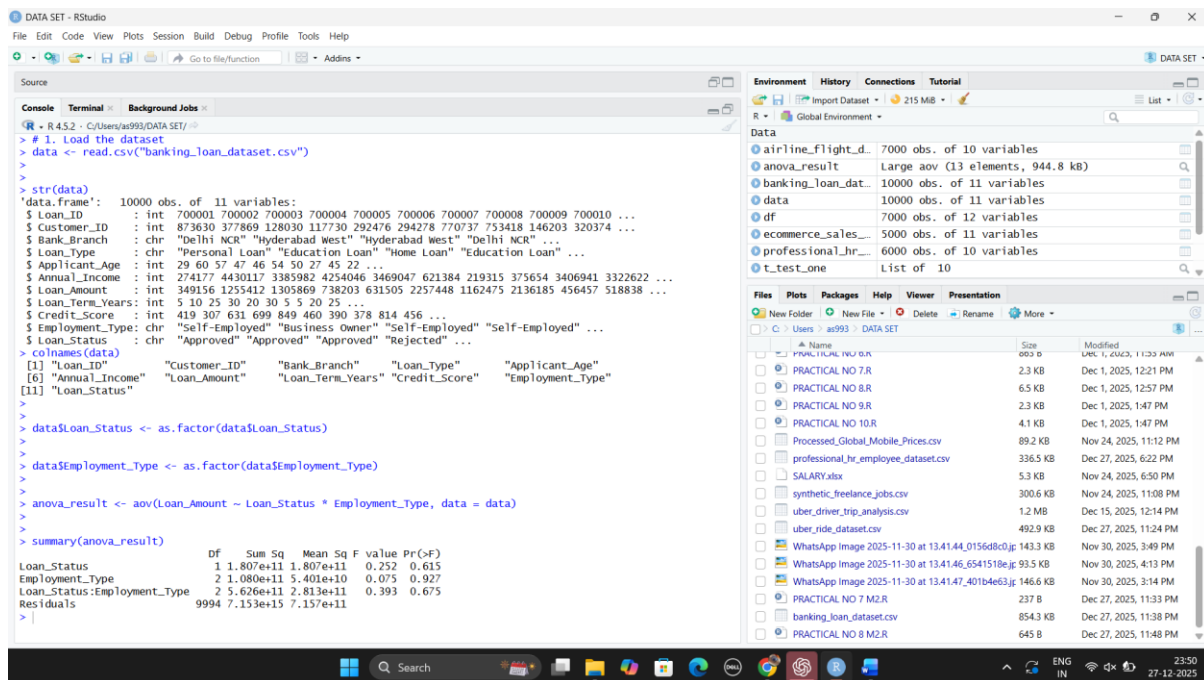


## PRACTICAL NO 8 MODULE 2

**AIM: Two-Way ANOVA using Car Price dataset**

**OUTPUT:-**



```
R - R 4.5.2 : C:/Users/as993/DATA SET/
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source Console Terminal Background Jobs
> # 1. Load the dataset
> data <- read.csv("banking_loan_dataset.csv")
>
> str(data)
'data.frame': 10000 obs. of 11 variables:
 $ Loan_ID      : int  700001 700002 700003 700004 700005 700006 700007 700008 700009 700010 ...
 $ Customer_ID  : int  873630 377869 128030 117730 292476 294278 770737 753418 146203 320374 ...
 $ Bank_Branch  : chr   "Delhi NCR" "Hyderabad West" "Hyderabad West" "Delhi NCR" ...
 $ Loan_Type    : chr   "Personal Loan" "Education Loan" "Home Loan" "Education Loan" ...
 $ Applicant_Age : int  29 60 57 47 46 54 50 27 45 22 ...
 $ Annual_Income : int  274177 4430117 3385982 4254046 3469047 621384 219315 375654 3406941 3322622 ...
 $ Loan_Amount  : int  349156 1255412 1305869 738203 631505 2257448 1162475 2136185 456457 518838 ...
 $ Loan_Term_Years : int  5 10 25 30 20 30 5 5 20 25 ...
 $ Credit_Score  : int  419 307 631 699 849 460 390 378 814 456 ...
 $ Employment_Type : chr   "Self-Employed" "Business Owner" "Self-Employed" "Self-Employed" ...
 $ Loan_Status   : chr   "Approved" "Approved" "Approved" "Rejected" ...
> colnames(data)
[1] "Loan_ID"      "Customer_ID"  "Bank_Branch"  "Loan_Type"    "Applicant_Age"
[6] "Annual_Income" "Loan_Amount"  "Loan_Term_Years" "Credit_Score" "Employment_Type"
[11] "Loan_Status"
>
> dataLoan_Status <- as.factor(data$Loan_Status)
>
> dataEmployment_Type <- as.factor(data$Employment_Type)
>
> anova_result <- aov(Loan_Amount ~ Loan_Status * Employment_Type, data = data)
>
> summary(anova_result)
              Df Sum Sq Mean Sq F value Pr(>F)
Loan_Status    1  1.807e+11 1.807e+11   0.252  0.615
Employment_Type 2  1.080e+11 5.401e+10   0.075  0.927
Loan_Status:Employment_Type 2  5.626e+11 2.813e+11   0.393  0.675
Residuals      9994  7.153e+15 7.157e+11
> |
```

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**ROLL NO: S116**